

C2-05 Python Automation for Analysts - Start Here

Begin here before PYA1

WHY THIS MODULE EXISTS

Course 1 Python teaches analysis. C2-05 teaches how to turn repeated analyst work into a small reliable pipeline: explicit inputs, reusable functions, defensive files/APIs, validation, logs, rerun safety and reproducible evidence.

EXACT ROUTE

PYA lesson -> matching P-PYA notebook section -> save genuine attempt + validation/failure evidence -> explain-back -> protected review -> close review -> changed retry. Repeat PYA1-PYA11. Then Weekly Reporting Mini-Lab -> Gate A. B/C remain protected.

CURRENT TARGET

Course references are refreshed to Python 3.14.7 and pandas 3.0.5. Live API access is optional; the deterministic local fixture is the assessed path. No paid service is required.

C2-05 prerequisites and tools

New automation concepts are still taught patiently from first principles

YOU SHOULD ALREADY KNOW

Basic Python syntax/functions and pandas DataFrame operations from Course 1. You should be able to read CSV, filter/group and perform a simple merge.

YOU NEED

Python, pandas, Requests and an Excel writer/reader engine for PYA5 if you run the Excel path. A notebook/VS Code editor is fine. Windows is fully acceptable.

RECOVERY

If base Python/pandas is weak, repair only that Course 1 skill. If a dependency/runtime is missing, record ENVIRONMENT BLOCKED. Do not replace hands-on evidence with screenshots from this book.

C2-05 mastery / Gate contract

No notebook-cell checkmarks and no percentage gaming

LESSON PASS

Clean runnable task + independent validation + one intentional failure/recovery + explain-back + changed retry. A happy-path notebook that only works in the current kernel is not mastery.

GATE A

Fresh integrated automation with files, transforms, joins, errors, logging, validation, rerun and manifest evidence.

B/C RETEST

Different domains/data/failure traps. Any unresolved critical file/schema/key/validation/rerun/reproducibility failure blocks pass.